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The component of velocity along vertical direction ($y$-axis) is $v_y = u_y + a_y t$

Since, $u_y = 0$, $a_y = g$, we get

$$v_y = gt \qquad \rightarrow (2.27)$$

Hence the velocity of the particle at any instant is

$$\vec{v} = u\hat{i} + gt\hat{j}$$

The speed of the particle at any instant $t$ is given by

$$\therefore \quad v = \sqrt{v_x^2 + v_y^2}$$
$$v = \sqrt{u^2 + g^2t^2}$$

**(4) Speed of the projectile when it hits the ground:** When the projectile hits the ground after initially thrown horizontally from the top of tower of height h, the time of flight is

$$t = \sqrt{\frac{2h}{g}}$$

The horizontal component velocity of the projectile remains the same i.e $v_x = u$

The vertical component velocity of the projectile at time $T$ is

$$v_y = gT = g\sqrt{\frac{2h}{g}} = \sqrt{2gh}$$

The speed of the particle when it reaches the ground is

$$v = \sqrt{u^2 + 2gh}$$

## 2.11.3 Projectile under an angular projection

This projectile motion takes place when the initial velocity is not horizontal, but at some angle with the vertical, as shown in Figure 2.42.

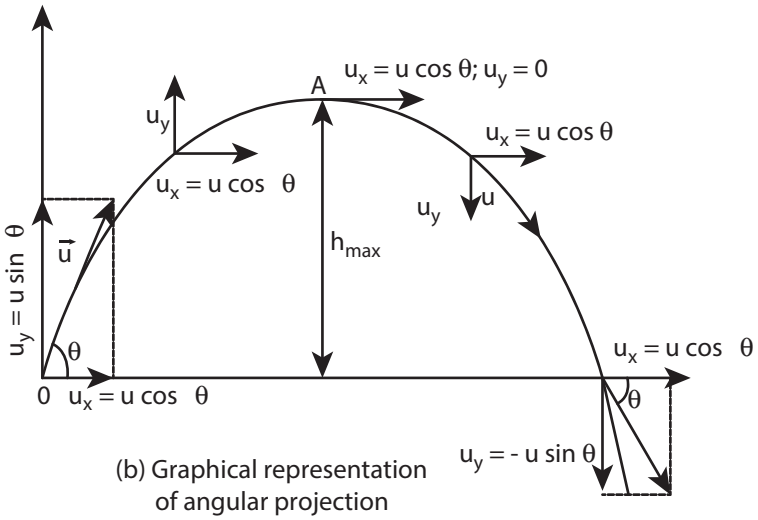
**(Oblique projectile)**

**Examples:**

- Water ejected out of a hose pipe held obliquely.
- Cannon fired in a battle ground.



(a) Water is ejected out of a pipe held obliquely




(b) Graphical representation of angular projection

**Figure 2.42.** Projectile motion